6.S981 Final Project Write-up

Conditional Generation from Few-shot Examples for Protein Design

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1 Abstract

Designing functional proteins with desired properties remains a central challenge in bioengineering, as the relationship between sequence and function is highly complex and experimental data is often limited. We propose a few-shot generative framework that conditions a generative multiple sequence alignment (MSA) protein language model on a small set of example sequences to synthesize novel proteins exhibiting desired features. The model extends a pretrained MSA Transformer with a diffusion-based generative objective and employs classifier-free guidance to enable controllable, context-dependent generation without relying on explicit functional labels or external scoring functions. Across a range of enzyme and deep mutational scanning benchmarks, we show that conditioning on functionally related sequences improves sequence-level fitness prediction and steers generation toward functionally coherent designs, demonstrating the effectiveness of in-context functional learning in low-data regimes.

2 Introduction

Designing proteins with desired functions is a central goal of modern biotechnology. From therapeutic proteins to industrial catalysts, navigating the immense protein space and discovering sequences with targeted structural and functional properties would transform protein engineering. However, this vision remains difficult to realize: known proteins occupy only a small fraction of the vast sequence space, the sequence-function landscape is highly rugged and poorly understood, and experimental synthesis and characterization are slow and costly. Therefore, generative models capable of inferring sequence patterns from functional examples and generalizing beyond the boundaries of observed data are essential for efficient and scalable protein design.

Recent advances in protein language models (PLMs) have shown that large-scale unsupervised learning can capture underlying principles of protein structure, stability, and evolution directly from sequences. However, to be used for specific downstream applications, most existing PLMs depend on fine-tuning on large labeled datasets. In domains with scarce data or incomplete labels, such as novel protein folds, understudied organisms, or emerging catalytic activities, these models struggle to adapt effectively. Developing PLMs that can generalize across diverse functional contexts from few-shot learning remains an open frontier.

Enzymes exemplify this challenge. They are nature's miracle of chemistry, driving nearly every essential chemical transformation of life. Biochemists categorize enzymatic functions using the Enzyme Commission (EC) system, which defines over 8,400 distinct catalyzed reaction classes in a hierarchy with four levels, from the broad classification to the most specific chemistry. Some leaf classes in the EC system contain only a handful of experimentally characterized sequences, reflecting the extreme data imbalance and sparsity of functional annotations. As a result, the diversity of classes and scarcity of labeled data make traditional fine-tuning approaches impractical for universal enzyme design. A learning framework that can generalize across enzyme classes and infer functional patterns from a few examples would therefore be highly valuable in this context.

To address this challenge, we introduce a few-shot generative framework for function-aware protein design. Rather than conditioning on discrete labels such as EC numbers, we provide the model with a small set of representative sequences that implicitly define a functional context. Our approach leverages a diffusion-based extension of the MSA Transformer, enabling the model to infer functional preferences from a few contextual sequences and generate novel proteins consistent with the preferences. We also explored using classifier-free guidance (CFG) to apply positive and negative preferences on the design targets. Conceptually, the model performs in-context functional learning

in the sequence space: given only a few examples with desired features, it infers the underlying functional manifold and generates novel sequences consistent with those features. We evaluate this framework on enzyme design as a case study; however, the methodology is broadly applicable to other protein domains where data are limited and functions are diverse, offering a general route toward data-efficient and function-conditioned protein design.

3 Related Work

De Novo and Generative Protein Design. Traditional protein design methods rely on physics-based modeling, such as Rosetta [1], and molecular dynamics simulations to optimize amino acid sequences for target structures, binding energies, and molecular interactions. While effective for small proteins and known folds, these approaches are computationally expensive. Recent deep learning advances have shifted towards data-driven de novo protein design, leveraging generative models such as VAEs, GANs, and autoregressive transformers. Diffusion-based frameworks like ProtDiff [2] and Chroma [3] extend this by generating 3D atomic coordinates for backbones. Most generative models require task-specific fine-tuning or explicit conditioning on labeled attributes (EC numbers, substrate classes, or turnover rates) to achieve functional specificity, limiting generalization to low-data or novel functional regimes. By enabling conditional generation from a few unlabeled examples, we hope to unify de novo generative design with few-shot functional conditioning.

Protein Language Models for Structure and Function. Large-scale unsupervised training on protein sequences has yielded models such as ESM2 [4], ProtT5 [5], and MSA Transformer [6], which capture evolutionary and structural constraints from sequences without explicit supervision. These PLMs achieve strong zero-shot performance for the prediction of downstream structure, contacts, and mutational effects, demonstrating that learning from sequence can encode meaningful biophysical information. Yet, their utility for generative design is constrained by the lack of task conditioning. Approaches such as ProtGen [7] and ProteinDT [8] adapt PLMs for conditioning generation by fine-tuning on labeled attributes like stability, yet it is challenging to adapt to novel or sparsely annotated enzyme families. We propose a few-shot conditioning approach that extracts implicit functional contexts from a small set of representative sequences, enabling PLMs to generalize to new enzymatic functions without explicit labels or retraining.

Machine Learning for Enzyme Function Classes. The intersection of enzyme engineering and machine learning reveals the relationship between enzyme function classes, protein 3D structures and peptide sequences. Representation learning of EC numbers like EC2Vec[9] embedded EC numbers into continuous vectors that capture their hierarchical relationships, improving enzyme-reaction classification. GraphEC[10] combined geometric graph learning on ESMFold structures with PLM embeddings to enhance EC-number and active-site prediction. To address biased and incomplete biochemical data, Okuno et al.[11] introduced a self-supervised domain-adaptation strategy that learns solely from positive enzyme–reaction pairs for turnover-rate and substrate-specificity prediction. Benchmarking studies across recent published EC number prediction models can be seen in Davoudi *et al.* [12] and Capela *et al.* [13]. Beyond predictive modeling, broader perspectives from Yang *et al.* [14] and Zhang *et al.* [15] surveyed how machine learning accelerates enzyme mining from existing databases and engineering beyond natural evolution, but emphasized persisting challenges of data scarcity and limited generalization across enzyme families. Therefore, our work moves toward universal generative modeling across low-data regimes of desired catalytic functions.

Few-shot In-context learning. In-context learning (ICL) enables large language models (LLMs) to adapt to new tasks from a handful of input–label examples without updating parameters [16]. Recent work extends few-shot ICL of general-purpose LLMs to biological sequences for antibody classification [17], suggesting that example-driven adaptation can improve data efficiency in biological tasks. However, because PLMs are trained on individual sequences rather than coherent passages, they do not naturally exhibit emergent cross-sequence reasoning. We therefore introduce an ICL-style conditioning module that explicitly aggregates information across few-shot examples and injects it into the generative process of a pre-trained PLM, enabling low-data protein design.

Controllable and Guided Generation. Controllable generation methods balance sample real-

ism against target-attribute fidelity. Classifier-based guidance steers diffusion using a separately trained attribute classifier [18], but depends on high-quality labeled data and robust classifiers. CFG [19] removes that dependency by mixing conditional and unconditional predictions with a single guidance weight, enabling smooth control without external classifiers. Given sparse labels in enzyme design, we therefore apply CFG to protein sequence generation to extrapolate from few-shot conditioning toward stronger functional alignment, while preserving plausibility from the unconditional PLM prior.

4 Methods

4.1 Dataset

Training data is curated from Pfam [20], a database that organizes protein sequences into multiple sequence alignments (MSAs) constructed using hidden Markov model profile (HMMs) derived from carefully curated seed alignments for each protein family. Families are further grouped into clans based on sequence and structural similarity.

To evaluate model generalization at both coarse-grained (clan) and fine-grained (family) levels, we adopt a hierarchical data-splitting strategy. First, we split the dataset at the clan level into training, validation, and test sets with an 8:1:1 ratio, while balancing the number of families per clan to account for the highly imbalanced distribution of families across clans. Families without clan annotations are included entirely in the training set. This results in approximately 23k MSAs in the clan-level training set, and 1k MSAs each for validation and test, spanning 80 clans. Within the clan-level training set, we further perform a family-level split, yielding 18k MSAs for training and 2k MSAs each for validation and test at the family level.

To equip the model with few-shot conditional generation capability, we intentionally restrict the depth of the MSA context during training. Specifically, for each training and validation instance, we randomly subsample the original MSA according to a negative binomial distribution over MSA depth, with an average depth of 5 and a maximum depth of 20. This encourages the model to infer sequence constraints from limited contextual information.

4.2 Training

Our training objective is to enable the model to learn implicit structural and functional constraints from a small set of context sequences and conditionally generate a compatible target sequence. To this end, we input a shallow MSA into the model, treating the top-row sequence as the reconstruction target and the remaining rows as context, and train the model in a self-supervised manner using a modified MSA Transformer architecture.

Formally, an MSA is represented as a matrix $x \in \mathbb{R}^{D \times L}$, where each row $x^{(i)}$, $i \in \{0, \ldots, D-1\}$ corresponds to an individual sequence and columns correspond to residue positions. Conditioned on the context rows $x^{(1:D-1)}$, the model aims to reconstruct the target sequence $x^{(0)}$.

While the original MSA Transformer employs axial attention to process MSAs, it uses a BERT-style masked language modeling objective and is not designed for generative modeling. We convert it into a diffusion-based language model by masking tokens according to a timestep-dependent noise schedule and injecting a learned timestep embedding into the token representations during denoising.

At each training step, the model samples a timestep t, corrupts the target sequence $x^{(0)}$ into $\tilde{x}^{(0)}$ by masking a set of residue positions $\mathcal{M}_t$ according to the timestep-dependent mask ratio, and then predicts the original tokens at masked positions. The training objective is:

$$\mathcal{L}(x; t, \theta) = \sum_{i \in \mathcal{M}_t} \log p(x_i^{(0)} \mid \tilde{x}^{(0)}, x^{(1:D-1)}, t, \theta). \tag{1}$$

The model consists of 12 transformer layers with 12 attention heads and an embedding dimension

of 768, totaling approximately 114M trainable parameters. We use 128 diffusion timesteps, a maximum sequence length of 1024, and a maximum of 16,384 tokens per batch. Training is conducted with a learning rate of 1×10^{-4}, gradient accumulation over 16 batches, dropout of 0.1, and no gradient clipping. Each model is trained for approximately 6 hours on a single NVIDIA L40S GPU, corresponding to 13k global optimization steps over 11 epochs. We train one model from scratch and another initialized from a pretrained MSA Transformer checkpoint.

4.3 Inference

At inference time, the model generates new sequences conditioned on a given set of context sequences by iteratively denoising an initially fully masked target sequence across diffusion timesteps.

To further enable property-guided generation, we implement classifier-free guidance. Given a conditioning task, we provide the model with a set of positive examples and a set of negative examples corresponding to the desired feature. Two denoising processes are run in parallel to obtain logits conditioned on each context, denoted as $\text{logits}_{\text{pos}}$ and $\text{logits}_{\text{neg}}$. The final logits are computed via linear extrapolation:

$$\text{logits} = \text{logits}_{\text{neg}} + \lambda \left(\text{logits}_{\text{pos}} - \text{logits}_{\text{neg}} \right), \tag{2}$$

where the guidance scale λ controls the strength and direction of conditioning. Setting $\lambda = 0$ corresponds to purely negative conditioning, $\lambda = 1$ to positive conditioning, and $\lambda > 1$ extrapolates further in the direction of the positive context.

5 Results

5.1 Conditional Enzyme Generation

Dataset. For the task of enzyme sequence generation under MSA context conditions of enzyme functions, we constructed two MSA datasets to assess the model's performance with limited contextual information.

- **Price-149 dataset.** This dataset originated from an experimentally validated collection of bacterial genes [21], and was curated as a challenging EC number prediction dataset that was wrongly annotated by automated methods [22], thus becoming a popular external test for EC number prediction models. For our task, we partitioned it into 27 groups based on shared first-three-level EC numbers (`x.y.z.*`), and performed MSA using the Python wrapper of Clustal Omega [23] to construct a context, from which the generative model designed new enzymes that match the same three-level EC number (`x.y.z`).
- **Rare Enzymes in UniProt dataset.** We assembled a UniProt-derived sequence pool by selecting reviewed sequences with EC numbers annotation and filtering out subunit entries to retain only full, single-chain enzymes [24]. From this filtered pool, we randomly selected 20 full EC numbers (`x.y.z.w`) that contain 10–30 sequences of length $<$1024 after MSA within each EC class, and these aligned sets served as contexts for generative models to design enzymes with the same first three levels of EC numbers (`x.y.z`).

Those datasets are designed to evaluate the models in realistic low-data scenarios: Price-149 provides an external experimental benchmark that is rarely seen in standard protein-model training corpora, while the curated UniProt subsets represent the common situations in early-stage functional characterization where limited experimental data exist but full database-scale diversity is not yet available.

Oracle Models. To computationally evaluate whether the sequences generated by our model successfully exhibit the intended enzymatic functions, we utilized three state-of-the-art EC number predictors. Guided by the EC-Bench benchmarking study [12], we selected `CLEAN` [25] and

`EnzBert` [26]. In addition, following another benchmarking study on protein language models (PLMs) for EC prediction [13], we adopted its recommendation on an `ec_number_prediction_ensemble`, which combines predictions from multiple fine-tuned PLMs as an ensemble. Because all three oracle models provide open-source code and released model weights, we are able to deploy and evaluate them on custom datasets locally. Because these models are trained on UniProt-derived datasets, we used the Price-149 dataset as an external evaluation set to quantify their reliability. We observed that all oracle models perform robustly on the first three levels of EC numbers (`x.y.z.*`), while the fourth digit (full EC classification) remains challenging (Figure 1(b)). As a result, we decided to keep our design target at the first three levels.

Experimental Setup. In our conditional enzyme generation workflow, an input MSA is fed into the language model as the context, and generated sequences are iteratively denoised by a Top-k sampler, which selects $k = 16$ masked tokens with the highest confidence to unmask until all tokens are revealed as 10 full sequences in an output MSA. After removing the alignment gaps and other special tokens, the oracle models predict the EC numbers of the cleaned sequences, whose first three levels are matched with those of the EC numbers of the context sequences to evaluate the design success rate (Figure 1(a)).

For each model checkpoint, we carried out this workflow and calculated the average design success rate over all EC numbers in the Price-149 (Figure 1(c)) and Rare Enzyme in UniProt datasets (Figure 1(d)), respectively, with all three oracle models.

We observe that more training epochs lead to overfitting and damage the generated results. As a consequence, indicated by a consistent trend across 3 oracles, the best models always appear at an epoch of less than 10, and begin to degrade during further training. The most severe performance decrease happened to the ESM-pretrained model on the Rare Enzyme in UniProt dataset, and the ESM-pretraining does not show a significant advantage over training from scratch.

We compared the best models (`toprow_esm-epoch7-step9008` for Price-149, and `toprow-esm-epoch5-step6756` for Rare Enzyme in UniProt) against a random mutation baseline to see the effect of in-context learning. We first calculated the maximum similarity of each generated sequence over all its contexts, and averaged it over all generated sequences for the same EC number target `x.y.z` as the averaged similarity $s_{x.y.z}$. Then a mutation rate of $1 - s_{x.y.z}$ was applied to 10 random sequences sampled from the original `x.y.z` context sequences to generate the random mutation sequences. The average design success rates predicted by 3 oracles for this random mutation baseline are shown in Figure 1(c)(d) by dashed horizontal lines with corresponding colors. While the improvement over the baseline is minor for the easier Rare Enzyme in UniProt dataset, where the success rate has reached 100% from the perspective of `ec_number_prediction_ensemble` oracle with little room for improvement, the best model on the Price-149 tasks surpasses the baseline to a certain extent.

5.2 Fitness Prediction with Classifier-Free Guidance

Experimental Setup. We evaluate fitness prediction performance using single-residue substitution datasets from ProteinGym [27]. We select 180 mutation sites, each drawn from a distinct protein sequence and experimental assay. For each site, we consider the complete set of 20 single-point amino acid substitutions covering all possible residue types at the mutated position. Mutated sequences are labeled as positive or negative based on their experimentally measured fitness scores, using thresholds specified by each dataset.

For each model checkpoint, we compute the token-level logits conditioned on the positive mutations ($\text{logits}_{\text{pos}}$) and negative mutations ($\text{logits}_{\text{neg}}$), respectively. Classifier-free guidance is applied to these logits to bias predictions toward either higher or lower fitness, controlled by the guidance scale λ, according to Equation 2. More specifically, we evaluate two guidance settings: a guidance scale of $+2.0$, which emphasizes positive fitness signals by extrapolating twice as far from negative toward positive, and a guidance scale of -1.0, which extrapolates the same difference toward lower fitness.

We compare the guided predictions against the ground-truth fitness measurements using two com-

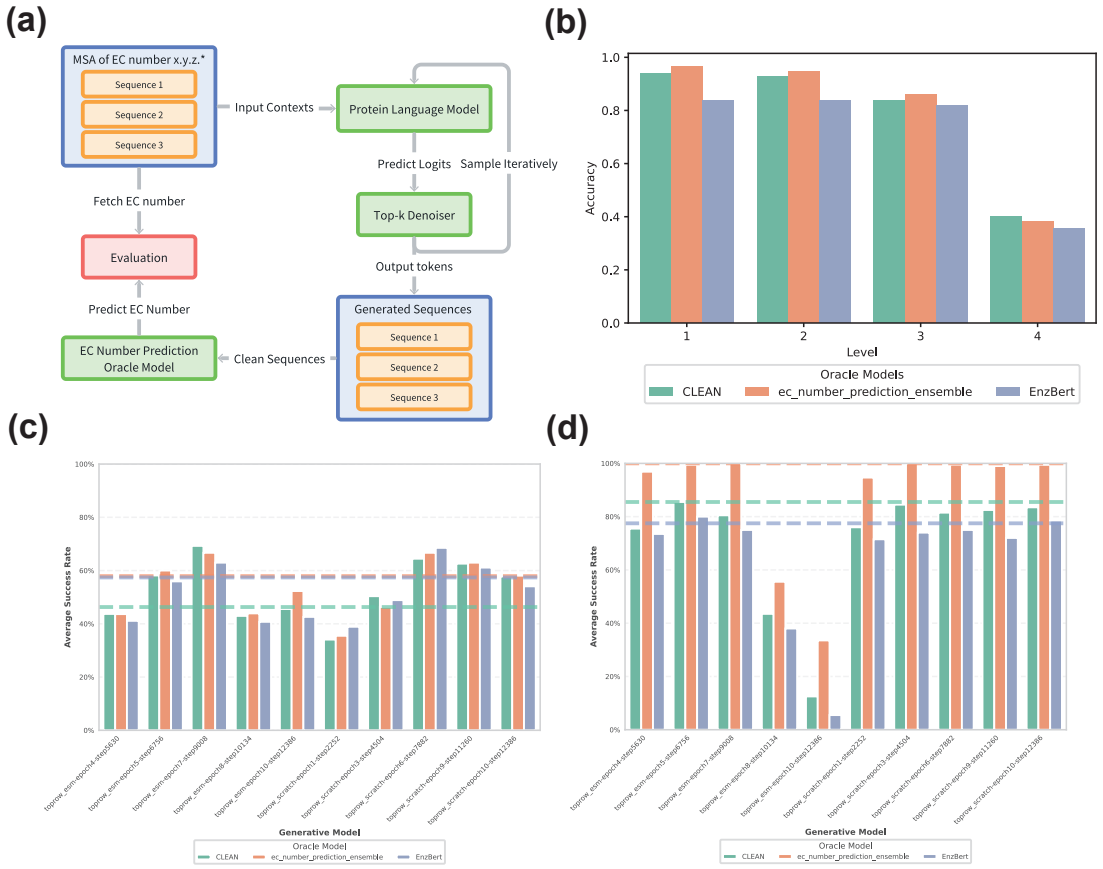


Figure 1: **Conditional enzyme generation with in-context learning from a few samples of a specific EC number.** (a) The generation and evaluation workflow. (b) The EC number prediction oracle models' success rate on the Price-149 dataset. (c) The average design success rate on the Price-149 dataset with Top-k sampling. (d) The average design success rate on the Rare Enzymes in UniProt dataset with Top-k sampling.

plementary metrics. First, we compute the *spearman correlation* between the normalized predicted probability distribution and the empirical fitness score distribution, capturing the monotonic alignment between model confidence and fitness. Second, *top-5 success rate* measures the fraction of experimentally positive mutations among the five most probable amino acid substitutions predicted by the model.

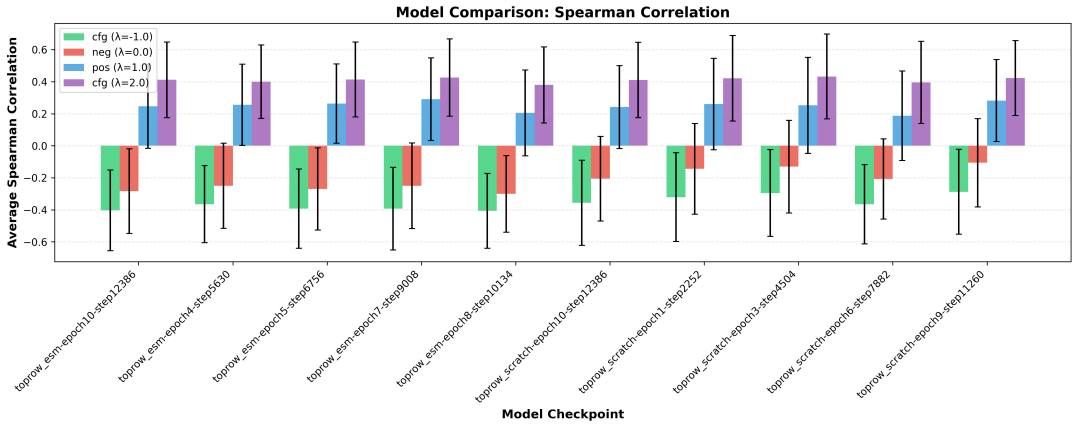


Figure 2: Spearman correlation between predicted probability distributions and ground-truth fitness scores across model checkpoints. Error bars denote standard deviation.

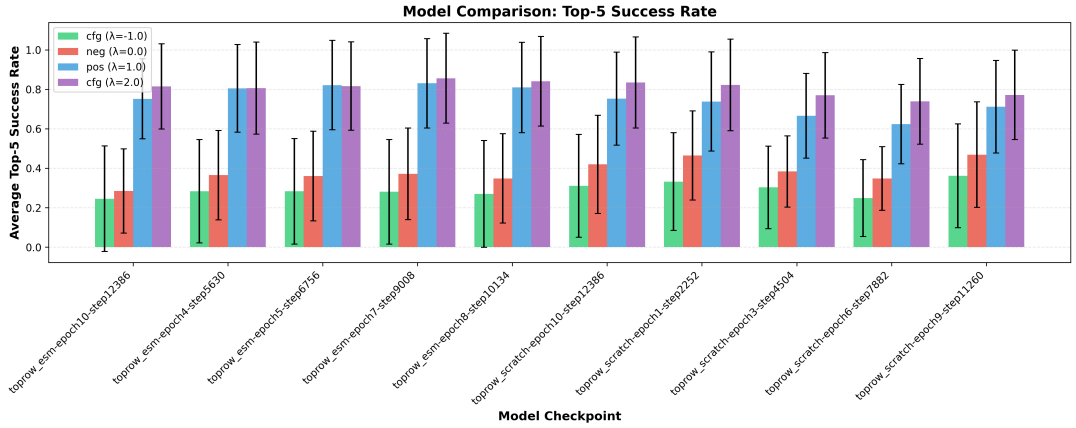


Figure 3: Top-5 success rate of fitness prediction across model checkpoints under different conditioning strategies. Error bars denote standard deviation.

Figures 2 and 3 compare fitness prediction performance across model checkpoints under conditional and classifier-free guided inference. First, we observe that token probability distributions conditioned on positive mutations exhibit a positive Spearman correlation with the empirical fitness scores and achieve substantially higher top-5 success rate. In contrast, predictions conditioned on negative mutations show negative Spearman correlation and low top-5 success rate. These results indicate that the model leverages contextual information to preferentially generate functionally similar mutations aligned with the provided conditioning examples.

Second, classifier-free guidance further amplifies this separation by explicitly extrapolating between positive and negative conditional distributions. As the guidance scale increases in magnitude, the Spearman correlation improves proportionally, aligning the ordering of candidate substitutions along the inferred fitness axis. In contrast, the improvements in Top-5 success rate are comparatively modest, suggesting that guidance predominantly reshapes the intermediate and tail regions of the probability distribution rather than dramatically altering the most confident predictions.

Finally, performance trends remain largely consistent across model checkpoints and are similar for models trained from scratch and those initialized from pretrained weights. This robustness suggests that the ability to extract fitness-relevant contextual signals is fairly easy for the model to acquire.

Together, these observations highlight conditional generative modeling and classifier-free guidance

as a simple yet effective mechanism for enhancing fitness-aware prediction by controllably reshaping conditional likelihoods, without requiring additional supervision or model retraining.

6 Discussion

These results highlight that conditional in-context learning can facilitate the generation of function-based proteins. By conditioning on a shallow MSA during inference, the model can infer signals related to the enzymatic function and use these implicit patterns to guide sequence generation and fitness prediction. This indicates that MSA Transformer does encode sufficient structural and evolutionary priors to extrapolate from small conditioning sets, without requiring explicit labels, additional supervision, or model retraining.

Across the Price-149 benchmarks and the UniProt-curated rate EC families, we were able to observe meaningful functional alignment in the generated sequences. In the low-data UniProt setting, where enzyme families contain only 10-30 known members, the best model achieves 50-90% functional success across the three independent oracle predictors (CLEAN, EnzBERT, PLM ensemble). This shows that the model can recreate functional sequence constraints from few-shot examples, even with a limited conditioning context. Although small, the model performance was still significantly above random mutation baselines in the more challenging Price-149 data set, in which enzyme classes are more sparsely represented in training. This suggests that when contextual sequences are close to the pretrained model's evolutionary priors, the functional generalization is the strongest. While the pretrained initialization offers a strong prior, we found that it did not consistently outperform models trained from scratch. This suggests that leveraging MSA-derived evolutionary information alone can be sufficient for functional generalization and that pretrained priors may require more tailored conditioning to provide additional benefit.

A central limitation of our current framework for conditional enzyme design is the reliance on MSA-derived contexts. Although MSA is the most practical and widely used approach for gathering enzyme sequences, it implicitly assumes that enzymes sharing functional properties also span a compact region in sequence space. In reality, the function space is far more entangled: enzymes that catalyze highly similar reactions may be distant in evolution with little detectable homology. As a result, MSA-based contexts can be heterogeneous, confusing, or even misleading for generative or predictive models trained under MSA contexts. Therefore, the next attempt of our few-shot enzyme generation should come from new methods of extracting functional similarity without depending on strict sequence similarity. This may require further development of pretraining objectives or model architectures to map diverse enzyme sequences into a shared functional manifold.

The inference strategy also seems to impact performance. We observed that the top-k sampling approach consistently improved success compared to confidence-based sampling and avoided excessively conservative generations of gap tokens. This indicates that being able to control diversity during generation can help preserve the underlying evolutionary constraints.

The single-site mutation experiments further support the emergence of function-aware model behavior. We observed that positive conditioning improves the rank-based correlation and top-5 classification of experimentally beneficial substitutions, while the negative conditioning predictably reverses these patterns. CFG can help further separate between high and low fitness substitutions without modifying the model parameters.

These results demonstrate that guided conditional generative modeling can provide a versatile tool to explore fitness landscapes. Further work includes incorporating structural priors (i.e., structure-conditioned diffusion guidance or embedding constraints) and MSA-conditioned log-likelihood-ratio fitness prediction. Additionally, we can further assess the novelty of the sequence diversity, beyond just the functional alignment.

7 Conclusion

We present a diffusion-based extension of the MSA Transformer that is capable of few-shot, function-conditioned protein generation using shallow evolutionary context rather than labeled supervision. Through experiments across diverse enzyme families, we show that the model can successfully generate sequences aligned to target EC functions, achieving up to 50-90% success. These findings highlight that we can learn and control function from a few biological examples through guided sampling. This represents potential for the shift towards generative in-context learning for protein engineering applications. Future work integrating structural constraints, explicitly optimizing for novelty and stability, and incorporating experimental validation to assess biological activity can further extend this framework. This line of work can ultimately help contribute towards flexible and generalizable protein design that can help accelerate enzyme discovery, especially across poorly characterized regions of protein sequence space.

8 Team Contributions

Zhilei Bei: Led the conceptualization of the few-shot conditional generation framework; curated and preprocessed the Pfam dataset for training; designed and implemented the model architecture; trained the models; conducted fitness prediction with classifier-free guidance experiments; and authored Section 1 (Abstract), Section 4 (Methods) and Section 5.2 (Fitness Prediction with Classifier-Free Guidance) of the manuscript.

Weiliang Luo: Proposed the application on conditional enzyme generation; preprocessed the UniProt and Price-149 dataset for model evaluation; helped with ideation of classifier-free guidance and implementation of sampling; deployed local EC number prediction models; conducted conditional enzyme generation experiments; and authored Section 2 (Introduction) and Section 5.1 (Conditional Enzyme Generation) of the manuscript.

Pari Latawa: Implemented random-mutation baseline and provided generated sequence sets for comparison; explored zero-shot fitness prediction for DMS datasets; and contributed Section 3 (Related Work), Section 6 (Discussion), and Section 7 (Conclusion).

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